Continuous Personalized Diffusion Model via Spinor-Component Forward Geometry

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Abstract

Existing conditional diffusion models typically inject condition information into the reverse-time denoising process, either as model inputs or through guidance. Although effective for conditional fidelity, this design burdens the denoiser when compact continuous signals, such as user preference ratios or mixture degrees, must support both image reconstruction and condition-wise separation, often degrading sample quality. In this work, we reinterpret this limitation not as a problem of condition *representation*, but as a problem of condition *placement*. We propose the **Continuous Personalized Diffusion Model (CPDM)**, which shifts the locus of conditioning from reverse-time signal injection to forward geometry formation by coupling bounded spinor-component coordinates with normalized image-space drift directions. The resulting condition-dependent forward drift forms condition-specific terminal geometry, from which continuous changes along the spinor-component coordinate are reflected as continuous visual transitions rather than endpoint interpolation. Experiments show that, even with a single scalar condition, CPDM achieves competitive performance compared with standard and higher-capacity reverse-conditioning baselines and supports continuous generation under compact scalar control. These results suggest forward geometry design as a practical alternative to conventional reverse-time conditioning for compact continuous personalization.

1 Introduction

Most existing conditional diffusion models treat condition information as a signal injected into the reverse-time denoising process, either as model inputs or through guidance [Dhariwal & Nichol, 2021, Ho & Salimans, 2022, Rombach et al., 2022]. This paradigm is effective for high-quality condition-aligned generation, but continuous personalization requires a different form of control. Signals such as user preference ratios or mixture degrees are intended to define a controllable path between conditions, rather than simply select one endpoint condition.

Interpolating between endpoint condition signals is a natural way to obtain such a path, but it does not necessarily produce meaningful intermediate samples: the reverse process still starts from a shared terminal space and must resolve both image reconstruction and condition-wise separation during denoising. Compact or low-dimensional signals provide a simpler continuous coordinate, but often degrade generation quality because the denoiser receives only a limited conditioning signal while being asked to separate different visual domains—an effect that worsens in inter-domain settings where endpoint distributions require different generative manifolds or geometries. These observations suggest that the core limitation lies not in condition representation, but in condition *placement*.

From this perspective, we propose the **Continuous Personalized Diffusion Model (CPDM)**, a forward-geometry-based conditional diffusion framework for compact continuous personalization.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Instead of using the condition only as a reverse-time steering signal, CPDM introduces a deterministic condition-dependent drift into the DDPM forward mean while preserving the standard variance schedule. Guided by the scale–direction template of the spinor formulation, this drift combines an explicit drift strength, bounded spin-component coordinates, and normalized image-space drift bases. Forward geometry organization thus becomes an explicit design problem—how far terminal states should be separated and which image-space direction should define that separation—while remaining fully compatible with the standard DDPM reverse machinery.

Experiments show that this minimal scalar design outperforms several standard and higher-capacity reverse-conditioning baselines and supports continuous generation, suggesting that *where* a condition acts in the diffusion trajectory can matter as much as *how* it is represented.

2 Related Work

2.1 Diffusion / Score-Based Generative Models

Diffusion models were originally introduced as generative processes based on nonequilibrium thermodynamics [Sohl-Dickstein et al., 2015], and DDPM later established a practical framework for sample generation through fixed forward noising and learned reverse Markov transitions [Ho et al., 2020]. Score-based SDE formulations further generalized this framework to continuous-time dynamics [Song et al., 2021b]. This shared Gaussian-prior formulation has remained a standard backbone not only for unconditional diffusion models, but also for conditional generation.

2.2 Reverse-Time Conditioning and Guidance

Conditional diffusion models have primarily evolved by injecting condition information into the denoiser or by modulating the reverse process through guidance [Dhariwal & Nichol, 2021, Ho & Salimans, 2022, Rombach et al., 2022]. These approaches provide strong conditional fidelity, but they primarily treat the condition as a steering signal for the reverse generation trajectory. In contrast, they do not explicitly make condition-specific geometry formation a central design component of the generative process.

2.3 Conditional Forward and Trajectory Shifting

Several recent studies have explored incorporating condition information into the forward process or trajectory-level dynamics. Shift-DDPM introduces condition-dependent trajectory shifts into the diffusion process [Zhang et al., 2023]. Instead of relying only on reverse-time denoising with a condition input, it assigns condition-specific diffusion trajectories by learning a shift predictor that determines the direction and magnitude of the trajectory shift. In this way, condition information affects the diffusion trajectory itself, and the quality of conditional generation depends on whether the learned shift direction provides a reliable separation between conditions.

CCDM addresses continuous conditional generation from a different angle [Ding et al., 2026]. For a continuous label y, CCDM constructs condition-dependent embeddings, including a short embedding used for U-Net conditioning and a long embedding used to define a label-dependent covariance matrix. This covariance structure is incorporated into the forward transition, reverse process, training objective, and sampling procedure. The denoising network predicts x_0 rather than the noise, and the training loss is weighted by the inverse of the condition-dependent covariance, so that image denoising is adapted to the local neighborhood of the continuous condition.

These approaches indicate that continuous or condition-specific generation can benefit from modifying the diffusion trajectory or forward statistics, rather than treating the condition only as an additional reverse-time input.

2.4 Our Positioning

We position CPDM as a minimal forward-organization approach for compact continuous personalization. Rather than increasing the capacity of reverse-time conditioning or introducing an auxiliary learned shift predictor, CPDM couples the condition coordinate directly to the DDPM forward mean. Compact spinor-component values act as bounded coefficients for normalized image-space drift bases, while the standard DDPM variance schedule is left unchanged. Consequently, the condition reorganizes the mean geometry of the forward distribution, rather than changing its covariance structure or the denoiser architecture. This restriction makes the forward design space explicit: the key variables are the strength, temporal schedule, and spatial direction of a deterministic drift.

3 Method

3.1 Standard DDPM Formulation: Forward and Reverse Processes

We consider data $x_0 \sim p_{\mathrm{data}}(x)$, a timestep $t \in \{1, \dots, T\}$, and Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$. Let $\{\beta_t\}_{t=1}^{T}$ denote the variance schedule of DDPM, and define

$$\alpha_t = 1 - \beta_t, \qquad \bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i. \tag{1}$$

The forward process of DDPM [Ho et al., 2020] is given by

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{2}$$

so the conditional distribution of x_t takes the closed-form Gaussian

$$q(x_t \mid x_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\, x_0,\ (1 - \bar{\alpha}_t) I\right). \tag{3}$$

In the reverse process, a learned denoiser $\epsilon_\theta(x_t, t)$—or equivalently an estimator $x_{0,\theta}(x_t, t)$—approximates $p_\theta(x_{t-1} \mid x_t)$ as a Gaussian transition, and samples are generated sequentially from $t = T$ to $t = 1$. Standard DDPM can thus be understood as a generative framework that learns the reverse mean μ_θ on top of a fixed Gaussian forward process.

The DDPM formulation above provides a general and tractable Gaussian forward process. However, in conditional generation, this shared forward process leaves two roles to the reverse denoiser: separating condition-dependent distributions and reconstructing fine image details. To reduce this burden, we seek a way to organize condition-dependent displacement within the forward process itself. This motivates a spinor-component construction, which provides bounded continuous coordinates together with a scale–direction template for forward geometry design.

3.2 Forward Geometry from Spinor Components

CPDM implements this idea by mapping the condition c to spin-component coordinates and coupling them to a deterministic condition-dependent forward drift. Rather than numerically integrating an additional latent-time spinor trajectory during diffusion, CPDM uses the spinor-induced observable coordinates in closed form. This formulation preserves the Gaussian tractability of $q(x_t \mid x_0, c)$ and keeps the model compatible with the standard DDPM reverse machinery, while allowing the condition to shape the forward terminal geometry before reverse denoising begins.

3.2.1 Spinor-Induced Condition Coordinates

We define a reduced complex two-component spinor as

$$\tilde{\psi}(\tau, c) = \begin{bmatrix} \alpha(\tau, c) \\ \beta(\tau, c) \end{bmatrix} \in \mathbb{C}^2. \tag{4}$$

Formally, we consider the following Schrödinger equation:

$$i \frac{d}{d\tau} \tilde{\psi}(\tau, c) = \hat{H}\, \tilde{\psi}(\tau, c), \qquad \hat{H} = B\, \hat{U}, \tag{5}$$

where B denotes the strength of the interaction and $\hat{U}$ denotes an operator that specifies a preferred direction in the spinor space. This relation provides the scale–direction template used to construct bounded spin-component values.

In the practical construction, condition dependence is represented by the normalized spinor components $\alpha(c)$ and $\beta(c)$. Let

$$\alpha(c) = f_1(c) + i f_2(c), \qquad \beta(c) = g_1(c) + i g_2(c), \tag{6}$$

where f_i and g_i, for $i = 1, 2$, are real-valued functions. Then the normalization condition is given by

$$|\alpha(c)|^2 + |\beta(c)|^2 = f_1(c)^2 + f_2(c)^2 + g_1(c)^2 + g_2(c)^2 = 1. \tag{7}$$

From this normalized spinor representation, we compute the expectation values of the spin components.[1] Thus, we define

$$s_i(c) = \tilde{\psi}(c)^\dagger \sigma_i \tilde{\psi}(c), \qquad i \in \{x, y, z\}. \tag{8}$$

In real-component form,

$$s_x(c) = 2\big(f_1(c)g_1(c) + f_2(c)g_2(c)\big), \tag{9}$$

$$s_y(c) = 2\big(f_2(c)g_1(c) - f_1(c)g_2(c)\big), \tag{10}$$

$$s_z(c) = \big(f_1(c)^2 + f_2(c)^2\big) - \big(g_1(c)^2 + g_2(c)^2\big). \tag{11}$$

In this work, condition information is encoded by the spin-component expectation values $(s_x(c), s_y(c), s_z(c))$, which lie in $[-1, +1]$ and serve as bounded coefficients for the condition-dependent drift. The spinor-component formulation therefore provides a normalized multi-axis coordinate interface, rather than an unstructured scalar condition.

3.2.2 From Spin Components to Multi-Axis Forward Drift

We translate spin-component expectation values into a condition-dependent drift applied to the DDPM forward process. Guided by the scale–direction structure of the Schrödinger-inspired relation, CPDM couples each spin component $s_i(c)$ to a fixed image-space drift direction $\hat{u}_i$. The general multi-axis drift is defined as

$$r_t^{\text{full}}(c) = A\, C_t \sum_{i \in \{x,y,z\}} s_i(c)\, \hat{u}_i, \tag{12}$$

where A controls the drift strength, C_t is the diffusion-time schedule, and each $\hat{u}_i \in \mathbb{R}^{H \times W \times 3}$ is a normalized image-space direction basis associated with the spin-component axis i. Thus, $s_i(c)$ acts as a signed coordinate coefficient for the corresponding image-space basis $\hat{u}_i$. As a result, the spinor-component formulation provides a structured basis for multi-axis and multi-style continuous control, where different semantic or stylistic directions can be assigned to different drift bases within a common bounded coordinate system.

In this work, we instantiate the simplest single-axis s_z case. This setting allows us to test whether a minimal scalar condition can improve generation performance through forward geometry formation and whether continuous transitions between two endpoint domains can be measured along the same coordinate. It therefore serves as a minimal single-axis validation of the broader multi-axis forward-drift interface. The scalar drift design used for this validation is described next.

3.3 Forward Geometry Design via Scalar Drift

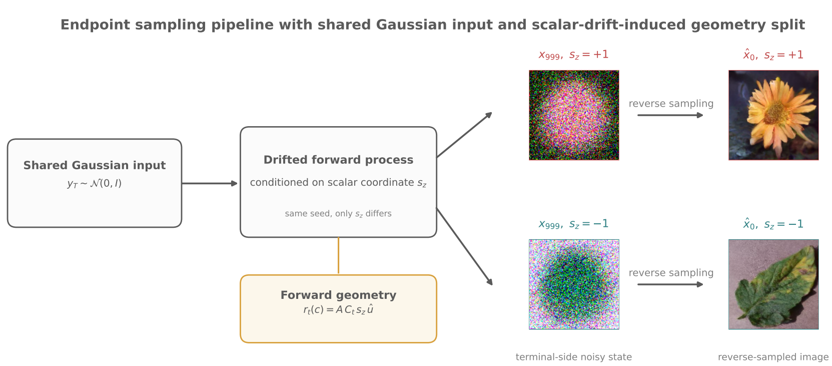


Figure 1: Practical scalar CPDM pipeline with shared Gaussian input and drift-induced forward geometry split. From a shared noise seed $y_T \sim \mathcal{N}(0, I)$, opposite scalar coordinates $s_z = +1$ and $s_z = -1$ induce separate terminal geometries, leading to visually distinct reverse-sampled outputs.

[1] Standard physical literature includes a factor of $1/2$ in the spin expectation value to reflect the nature of half-integer spin fermions. In the present work, however, this factor is omitted to simplify the notation.

Our study focuses on the scalar case, where the condition-dependent drift is defined as

$$r_t(c) = A\, C_t\, s_z(c)\, \hat{u}_z, \tag{13}$$

where A controls the drift strength, C_t controls temporal accumulation, and $\hat{u}_z$ is the fixed image-space drift basis defined in Section 3.2.2. To avoid scale-driven differences across drift bases, each $\hat{u}$ basis is globally rescaled to have an average pixel-wise ℓ_2 norm of 0.6 over its channel dimension, preserving the spatial structure of the basis while controlling the overall drift energy scale. We do not learn $\hat{u_z}$, because making the drift basis trainable would turn the forward perturbation itself into a moving target and would no longer correspond to the fixed-forward noise-prediction setting used by DDPM. For continuous conditioning, we directly use the spinor-induced coordinate on the z-axis.

$$s_z(c) = |\alpha(c)|^2 - |\beta(c)|^2. \tag{14}$$

Because the spinor is normalized as $|\alpha(c)|^2 + |\beta(c)|^2 = 1$, we have $s_z(c) \in [-1, 1]$. The endpoints $s_z = +1$ and $s_z = -1$ correspond to the two endpoint domains, while $s_z = 0$ represents their balanced midpoint. Thus, $s_z(c)$ is not used as a binary label, but as a continuous signed coordinate that directly controls the condition-dependent forward geometry.

Adding this scalar drift to the standard DDPM forward process gives

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \big(\epsilon + r_t(c)\big), \qquad \epsilon \sim \mathcal{N}(0, I). \tag{15}$$

Accordingly, the conditional forward distribution becomes

$$q(x_t \mid x_0, c) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, r_t(c),\ (1-\bar{\alpha}_t)I\big)\,. \tag{16}$$

Taking expectation over ϵ, the terminal mean is

$$\mathbb{E}_\epsilon[x_T \mid x_0, c] = \sqrt{\bar{\alpha}_T}\, x_0 + \sqrt{1-\bar{\alpha}_T}\, r_T(c), \tag{17}$$

so the condition-dependent terminal shift is $\sqrt{1-\bar{\alpha}_T}\, r_T(c)$. Thus, CPDM preserves the variance structure of the Gaussian forward process while reorganizing the terminal geometry by shifting only the condition-dependent mean coordinate.

The time schedule $C_t \in [0, 1]$ controls how the drift accumulates along the diffusion time axis, with $C_0 = 0$ and $C_T = 1$. Together with the drift strength A and the image-space basis $\hat{u}_z$, it determines how the scalar coordinate $s_z(c)$ is translated into controllable terminal geometry.

3.4 η-Target Learning Objective

CPDM is trained on endpoint boundary bands rather than interior mixture coordinates, so continuous responses at $s_z \in (-1, +1)$ are treated as generalization from real endpoint-domain data rather than as a consequence of explicit interpolation or mixture-target supervision.

3.4.1 Base CPDM

We define the drift-augmented target as $\eta_t(s_z) = \epsilon + r_t(s_z)$ and train the network to predict this combined noise-drift quantity. Base CPDM samples s_z uniformly from the boundary band $[-1.0, -0.95] \cup [0.95, 1.0]$, and minimizes the following η-target loss:

$$\mathcal{L}_{\text{main}} = \mathbb{E}_{x_0,\, \epsilon,\, t,\, s_z \sim q_{\text{bd}}} \Big[\|\hat{\eta}_\theta(x_t(s_z),\, t,\, s_z) - (\epsilon + r_t(s_z))\|_2^2 \Big]. \tag{18}$$

This objective exposes the model not only to the exact endpoints, but also to nearby continuous values of s_z, encouraging η-prediction to remain stable under small coordinate changes.

3.4.2 Continuous CPDM

To improve local smoothness along the scalar control axis, Continuous CPDM preserves the endpoint-band objective of Base CPDM and adds a pairwise consistency regularizer between nearby scalar coordinates. Specifically, we construct nearby condition pairs (s_1, s_2) that share the same x_0, timestep t, and noise ϵ, with both coordinates lying in the same boundary band defined in Section 3.4.1 and $|s_1 - s_2| \in [0.01, 0.05]$. We then introduce the pair consistency regularizer:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}\left[\frac{\big\| \big(\hat{\eta}_\theta(x_t(s_2), t, s_2) - \hat{\eta}_\theta(x_t(s_1), t, s_1)\big) - \big(r_t(s_2) - r_t(s_1)\big) \big\|_2^2}{\|r_t(s_2) - r_t(s_1)\|_2^2 + \varepsilon_{\text{pair}}} \right]. \tag{19}$$

and define the final objective as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{main}} + \lambda \, \mathcal{L}_{\text{diff}}. \tag{20}$$

Here, $\mathcal{L}_{\text{diff}}$ enforces local consistency around the endpoint anchors and stabilizes the model response to small scalar-coordinate changes.

3.5 Reverse Sampling and KL Alignment in the DDPM-Equivalent y-World

The forward process of CPDM shifts only the conditional mean through a deterministic drift:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\,\big(\epsilon + r_t(c)\big). \tag{21}$$

Define $m_t(c) := \sqrt{1-\bar{\alpha}_t}\, r_t(c)$, and the translated variable

$$y_t := x_t - m_t(c). \tag{22}$$

Substituting gives

$$y_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \tag{23}$$

which exactly recovers the standard DDPM forward form. Therefore, the drift-augmented forward process of CPDM is aligned with standard DDPM in the translated y-world. Since this translation has unit Jacobian for each fixed t and c, the corresponding KL terms can be analyzed in the standard DDPM y-world, and CPDM reuses the standard DDPM reverse machinery without introducing a new reverse-time Gaussian family. The full derivation and sampling algorithm are provided in Appendix A.

4 Experiments

4.1 Experimental Settings

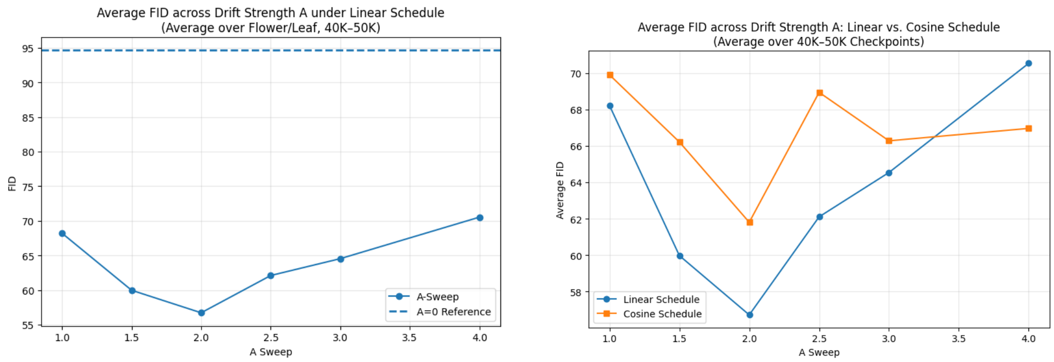
We use the Plant Disease Leaf dataset [Hughes & Salathé, 2015], the Oxford Flowers 102 dataset [Nilsback & Zisserman, 2008], and the CelebA-HQ 128×128 dataset [JJuOn, 2022] with 10,000 male and 10,000 female images. The Leaf/Flower setting is used as a coarse inter-domain conditional generation task, whereas CelebA-HQ gender is used as an intra-domain attribute-control task within the same face domain. Since CelebA-HQ involves a more subtle attribute-level separation, the detailed CelebA-HQ quantitative results are provided in Appendix E.4. All models are trained with a learning rate of 1×10^{-4} and a gradient clipping threshold of 3. For ablation studies, we use 1,000 generated samples per domain. For the main comparison with baseline models, we generate 5,000 samples per domain under training seeds 777 and 2026. For a fair comparison, all DDPM-based baselines and CPDM variants share the same denoising backbone.

4.2 Scalar CPDM: Forward Geometry Design Ablations

In this section, we analyze how changes in forward geometry design affect generation performance in scalar CPDM, while keeping the same compact scalar condition s_z. To this end, we conduct ablation studies by varying the drift strength A, the drift time schedule C_t, and the drift basis $\hat{u}_z$.

4.2.1 Effect of Drift Strength A

All drift-on settings achieved lower FID than the drift-off baseline. The best performance was observed around $A = 2$, while both smaller and larger drift strengths degraded performance. Under the linear schedule, $A = 2$ achieved the lowest Avg FID of 56.72. Comparing linear and τ-cosine schedules further shows that both time-dependent schedules reached their optimum near $A = 2$, while τ-cosine exhibited more gradual degradation at larger A. This suggests that drift strength controls the balance between terminal separation and reverse reconstruction. Under our normalization $\|\hat{u}_z\|_{\text{avg}} = 0.6$, $A = 2$ corresponds to a maximum average drift norm of approximately 1.2, where the best performance was observed.



(a) Linear schedule with drift-off reference.

(b) Linear vs. τ-cosine schedules.

Figure 2: Effect of drift strength A on average FID. FID is averaged over Flower/Leaf checkpoints from 40K to 50K. Left: the linear schedule consistently improves over the drift-off reference and achieves the best result near $A = 2$. Right: both linear and τ-cosine schedules show an optimum around $A = 2$, while larger drift strengths degrade generation quality.

4.2.2 Effect of Drift Time Schedule

To isolate the effect of temporal drift design, we compare linear, τ-cosine, and step-constant time schedules with matched terminal scale. Linear achieved the best peak performance with an Avg FID of **56.72**, outperforming τ-cosine (**61.80**) and constant-time scheduling (**72.32**). The step-constant schedule performs worst, indicating that explicit temporal drift accumulation matters for forward geometry formation: matching the terminal magnitude alone is insufficient, and the intermediate trajectory also affects reconstruction quality. Detailed definitions of the drift time schedules, their induced mean-shift trajectories, and the schedule comparison summary are provided in Appendix D.

4.2.3 Effect of Drift Basis $\hat{u}_z$

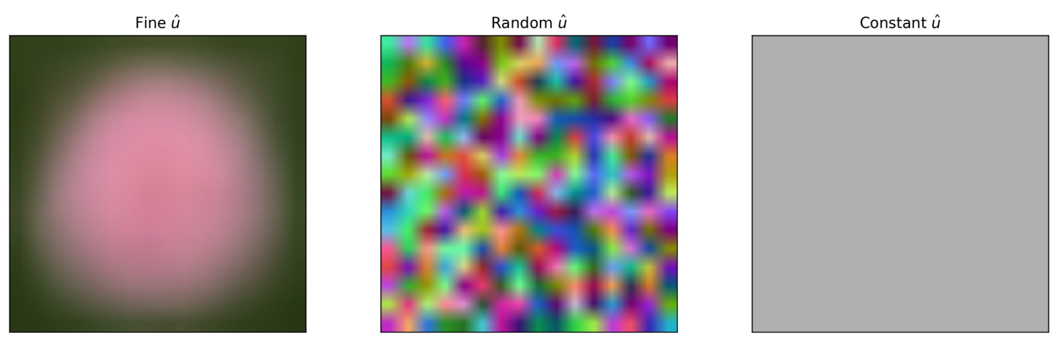


Figure 3: Visualization of image-space drift bases $\hat{u}$ used in the drift-basis ablation. All bases are globally rescaled to the same average per-pixel norm, so the comparison isolates the effect of spatial structure rather than total drift magnitude.

To further examine whether the performance gain comes from the spatial structure of the forward drift direction, we compare different choices of the image-space basis $\hat{u}$ while fixing the drift strength and time schedule. The fine basis is constructed from the difference between Flower and Leaf prototypes, each obtained by averaging 512 training images from the corresponding endpoint domain. The random basis is sampled from a random image-space direction, and the constant basis is a spatially uniform direction vector. All bases are globally rescaled to the same average per-pixel ℓ_2 norm of 0.6, so the comparison isolates spatial structure rather than drift magnitude.

With $A = 2$ and a linear schedule fixed, we compare fine, random, and constant $\hat{u}$ bases. Fine achieves the best Avg FID of 56.72, while random $\hat{u}$ still improves over the drift-off reference (88.25 vs. 94.61), and constant $\hat{u}$ slightly underperforms it (95.66). These results show that CPDM benefits from spatially structured forward geometry. Fine bases provide the most effective structured directions, and even random $\hat{u}$ can improve over the drift-off setting. In contrast, the spatially uniform constant basis fails to provide the same benefit, suggesting that the proposed drift is not merely a nonzero mean shift but depends on how conditional energy is spatially allocated in the forward process.

4.3 Comparison with Standard and Stronger Baselines on Leaf and Flower

Based on the ablation results in Section 4.2, we fix the selected scalar CPDM configuration and compare it with standard and stronger conditional baselines on the Leaf/Flower setting. This comparison evaluates whether the proposed forward-geometry design can remain competitive against higher-dimensional condition embeddings and stronger reverse-time predictors. We compare against Shift-DDPM [Zhang et al., 2023], CLIP-based conditioning [Radford et al., 2021], and LLaVA-caption conditioning [Liu et al., 2023].

Table 1: Comparison with baselines on the Leaf and Flower datasets. Avg FID is averaged over the two endpoint domains. FID Gap denotes the absolute Flower–Leaf FID difference at the reported best checkpoint, measuring domain-wise imbalance. Lower is better.

Model	Avg FID↓	Avg KID↓	FID Gap ↓	Best Step
Standard baselines				
Shift-DDPM	134.27 ± 10.31	$0.1315 \pm .0025$	31.05 ± 8.33	35K
Cond. DDPM (one-hot)	83.76 ± 9.30	$0.0772 \pm .0022$	48.87 ± 30.44	40K
Base CPDM (ours)	$\mathbf{49.68 \pm 5.96}$	$\mathbf{0.0412 \pm .0016}$	27.20 ± 15.59	50K
Stronger baselines				
Shift-DDPM++	126.94 ± 4.39	$0.1211 \pm .0024$	19.61 ± 19.12	35K
Jointly-learned 256D	84.60 ± 2.76	$0.0767 \pm .0023$	39.99 ± 31.67	40K
CLIP 512D image emb.	$\mathbf{29.74 \pm 0.27}$	$\mathbf{0.0184 \pm .0009}$	19.70 ± 0.67	50K
LLaVA-1.5 + CLIP text emb.	76.01 ± 0.02	$0.0686 \pm .0005$	36.17 ± 19.80	40K

Here, FID Gap measures the absolute difference between Flower-domain FID and Leaf-domain FID at the reported best checkpoint, so it reflects whether a model improves average quality by favoring only one endpoint domain.

Base CPDM substantially improves over standard baselines and also outperforms the larger shift-predictor variant Shift-DDPM++, jointly-learned 256D conditioning, and LLaVA-caption + CLIP text embedding, despite using only a compact scalar condition and a fixed drift basis. This suggests that generation quality is not determined only by the richness of the condition input; rather, forward geometry organization can provide an effective generative structure even when the condition signal is reduced to a single scalar coordinate. However, the CLIP image-embedding baseline remains the strongest model, indicating that direct image-level semantic features still provide the highest-capacity conditioning signal in this setting.

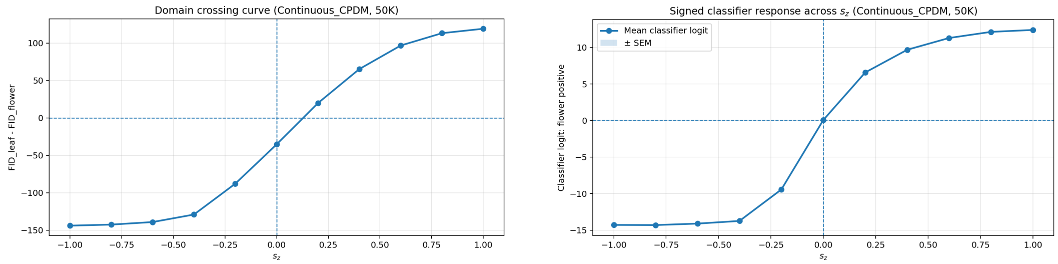
4.4 Quantitative Comparison and Scalar Traversal

4.4.1 Endpoint Comparison: Base CPDM vs. Continuous CPDM

Continuous CPDM achieves an Avg FID of 45.29 on the Leaf/Flower setting under Training seed 777, compared to 43.71 for Base CPDM. While endpoint-domain FID scores remain comparable, Continuous CPDM yields more stable non-collapsed responses along the scalar control axis. A direct comparison and s_z trajectory with Base CPDM under the identical seed is provided in Appendix F.

4.4.2 Monotonic Traversal Analysis

We evaluate Continuous CPDM along $s_z \in \{-1.0, -0.8, \ldots, 0.8, 1.0\}$ using two complementary signals: the signed domain-wise FID difference $\mathrm{FID}_{Flower}(s_z) - \mathrm{FID}_{Leaf}(s_z)$ and a probabilistic endpoint classifier score. These measurements are used to examine whether the scalar coordinate produces an ordered traversal between the two endpoint domains.



(a) Signed FID difference vs. s_z (b) Endpoint classifier response vs. s_z

Figure 4: Traversal analysis for Continuous CPDM across $s_z \in \{-1.0, -0.8, \ldots, 0.8, 1.0\}$. **Left**: Signed FID difference $\mathrm{FID}_{Flower}(s_z) - \mathrm{FID}_{Leaf}(s_z)$ computed using 1,000 generated samples per coordinate. **Right**: Endpoint probability estimated by a classifier trained only on real endpoint images and applied to the same 1,000 generated samples; the classifier is used solely for evaluation.

Although Continuous CPDM incurs a small endpoint-fidelity trade-off relative to Base CPDM, the monotonic behavior of both the signed FID-difference curve and the probabilistic classifier response provides direct evidence that s_z acts as an ordered distributional control axis. Rather than producing unordered mixtures or endpoint collapse, changing s_z progressively shifts the generated distribution from one endpoint domain toward the other. Nevertheless, this shift is not uniformly smooth across the full interval: the dominant distributional transition occurs around $s_z \in [-0.4, 0.4]$, indicating that CPDM learns an ordered but relatively sharp crossing region between the endpoint distributions.

5 Limitations

Our main empirical results focus on inter-domain generation, where a prototype-difference drift basis provides a clear separation direction between visually distinct endpoint domains. However, the design of the image-space drift basis $\hat{u}$ becomes more challenging in intra-domain attribute settings. In CelebA-HQ gender control, a raw difference-based prototype direction was not sufficient to surpass high-capacity semantic conditioning baselines such as LLaVA-caption + CLIP text embedding and CLIP image embedding. This suggests that the effectiveness of CPDM depends not only on introducing a nonzero forward drift, but also on how $\hat{u}$ is aligned with the semantic and spatial structure of the target condition. A more detailed characteristic analysis of this drift-basis behavior is provided in Appendix E.

These observations indicate that extending CPDM from coarse inter-domain separation to subtle intra-domain control requires more systematic drift-basis design. CPDM has also not yet been validated on larger-scale benchmarks such as ImageNet. These extensions remain future work.

6 Conclusion

In this work, we proposed CPDM, a forward-geometry-based conditional diffusion framework that shifts condition-wise separation from reverse-time conditioning to the forward process. The results show that conditional generation quality is not determined only by the richness of the condition representation: by reorganizing the forward mean geometry, CPDM can reduce the denoising burden and outperform several standard and stronger reverse-conditioning baselines even under compact scalar control. Beyond endpoint quality, CPDM also supports ordered scalar traversal without explicit interior-mixture supervision or endpoint interpolation, suggesting that condition-reorganized forward space can serve as a practical design axis for more continuous and controllable personalized image generation.

Future work will extend CPDM along two directions. First, we will investigate automatic construction of the drift basis $\hat{u}$ by using CLIP image-embedding space to select or mix candidate bases such as $\hat{u}_{\text{diff\_raw}}$, $\hat{u}_{\text{diff\_lowpass}}$, $\hat{u}_{\text{edge}}$, and $\hat{u}_{\text{common}}$, enabling more transferable drift design across inter-domain and intra-domain settings. Second, we will extend the scalar coordinate to multi-axis personalization by aligning (s_x, s_y, s_z) and $(\hat{u}_x, \hat{u}_y, \hat{u}_z)$ with distinct semantic axes such as gender, bangs, and eyeglasses.

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A KL/ELBO Alignment in the DDPM-Equivalent y-World

A.1 Setup and assumptions

In this appendix, we provide a rigorous justification that the proposed drift-augmented forward process remains variationally aligned with standard DDPM after translation to the DDPM-equivalent y-world.

For each fixed condition c, we assume that the drift term $r_t(c)$ is deterministic and depends only on the condition and time index, not on x_t, x_0, or ϵ. For notational convenience, define the corresponding mean-shift term

$$m_t(c) := \sqrt{1-\bar{\alpha}_t}\, r_t(c). \tag{24}$$

Throughout this appendix, $r_t(c)$ denotes the drift in the eta-target or noise-space parameterization, whereas $m_t(c)$ denotes the actual mean shift in x_t-space.

Then the CPDM forward process can be written as

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + m_t(c) + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{25}$$

Equivalently,

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\big(\epsilon + r_t(c)\big). \tag{26}$$

We also extend the cumulative drift schedule by convention with

$$C_0 := 0. \tag{27}$$

Since the drift is defined as

$$r_t(c) = AC_t s_z(c)\hat{u}_z \tag{28}$$

in the scalar case, it follows that

$$r_0(c) = 0 \quad \text{and} \quad m_0(c) = \sqrt{1-\bar{\alpha}_0}\, r_0(c) = 0. \tag{29}$$

Therefore,

$$x_0 = y_0 \tag{30}$$

holds exactly, which will be important for the reconstruction term in the ELBO.

To match the practical sampling procedure

$$y_T \sim \mathcal{N}(0, I), \qquad x_T = y_T + m_T(c), \tag{31}$$

the generative factorization in x-space should be written as

$$p_\theta(x_{0:T} \mid c) = p(x_T \mid c) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t, c), \tag{32}$$

with the conditional prior

$$p(x_T \mid c) = \mathcal{N}\big(m_T(c), I\big). \tag{33}$$

This is the x-space counterpart of the standard Gaussian prior

$$p(y_T) = \mathcal{N}(0, I) \tag{34}$$

in the translated y-world.

Thus, the prior is unconditional in the translated y-world, but condition-dependent in x-space because the practical sampler initializes x_T by translating y_T with $m_T(c)$.

A.2 Forward marginal in the translated y-world

Define the translated variable

$$y_t := x_t - m_t(c). \tag{35}$$

Substituting the CPDM forward expression gives

$$y_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon. \tag{36}$$

Hence the conditional forward marginal in y-space is

$$q_y(y_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\, (1-\bar{\alpha}_t)I\big), \tag{37}$$

which is exactly the standard DDPM forward marginal.

By contrast, the forward marginal in x-space is

$$q_x(x_t \mid x_0, c) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0 + m_t(c),\, (1-\bar{\alpha}_t)I\big). \tag{38}$$

Thus CPDM preserves the Gaussian closed form of DDPM, while shifting only the conditional mean geometry through $m_t(c)$.

A.3 KL invariance under condition-fixed translation

For each fixed t and c, consider the translation map

$$T_t^{(c)} : x_t \mapsto y_t = x_t - m_t(c). \tag{39}$$

Because $m_t(c)$ is deterministic, this map is bijective, with inverse

$$\big(T_t^{(c)}\big)^{-1}(y_t) = y_t + m_t(c). \tag{40}$$

Its Jacobian is the identity:

$$J_{T_t^{(c)}} = I, \qquad \Big|\det J_{T_t^{(c)}}\Big| = 1. \tag{41}$$

Therefore, for any two densities q_x and p_x on x_t, and their pushforwards q_y and p_y under $T_t^{(c)}$, KL divergence is preserved:

$$D_{\mathrm{KL}}(q_x \| p_x) = D_{\mathrm{KL}}(q_y \| p_y). \tag{42}$$

This is the basic reason why the KL terms of CPDM in x-space can be analyzed through the standard DDPM terms in the translated y-world.

A.4 Terminal KL term

At the terminal step,

$$q_x(x_T \mid x_0, c) = \mathcal{N}\big(\sqrt{\bar{\alpha}_T}\, x_0 + m_T(c),\, (1-\bar{\alpha}_T)I\big), \tag{43}$$

while the x-space prior is

$$p(x_T \mid c) = \mathcal{N}\big(m_T(c), I\big). \tag{44}$$

Now apply the translation

$$y_T = x_T - m_T(c). \tag{45}$$

Then,

$$q_y(y_T \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_T}\, x_0,\, (1-\bar{\alpha}_T)I\big), \tag{46}$$

and

$$p(y_T) = \mathcal{N}(0, I). \tag{47}$$

Since the translation has Jacobian 1,

$$D_{\mathrm{KL}}(q_x(x_T \mid x_0, c) \,\|\, p(x_T \mid c)) = D_{\mathrm{KL}}(q_y(y_T \mid x_0) \,\|\, p(y_T))\,. \tag{48}$$

Therefore, the terminal KL term of CPDM is exactly the standard DDPM terminal KL term in the translated y-world.

A.5 Intermediate posterior correspondence

We next show that the intermediate forward posterior in x-space is a translated version of the standard DDPM posterior in y-space.

Define

$$y_t = x_t - m_t(c), \qquad y_{t-1} = x_{t-1} - m_{t-1}(c). \tag{49}$$

Note that the shifts at t and $t-1$ are generally different, i.e.,

$$m_t(c) \neq m_{t-1}(c). \tag{50}$$

Therefore, the posterior equivalence should be justified through the joint density, rather than by a single-step informal translation argument.

Consider the joint transformation

$$(x_{t-1}, x_t) \mapsto (y_{t-1}, y_t) = \big(x_{t-1} - m_{t-1}(c),\, x_t - m_t(c)\big). \tag{51}$$

Its Jacobian is block-diagonal:

$$J = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}, \qquad |\det J| = 1. \tag{52}$$

Hence,

$$q_x(x_{t-1}, x_t \mid x_0, c) = q_y(y_{t-1}, y_t \mid x_0). \tag{53}$$

Similarly,

$$q_x(x_t \mid x_0, c) = q_y(y_t \mid x_0). \tag{54}$$

Therefore,

$$\begin{aligned} q_x(x_{t-1} \mid x_t, x_0, c) &= \frac{q_x(x_{t-1}, x_t \mid x_0, c)}{q_x(x_t \mid x_0, c)} \\ &= \frac{q_y(y_{t-1}, y_t \mid x_0)}{q_y(y_t \mid x_0)} \\ &= q_y(y_{t-1} \mid y_t, x_0). \end{aligned} \tag{55}$$

Since the posterior in the y-world is exactly the standard DDPM posterior,

$$q_y(y_{t-1} \mid y_t, x_0) = \mathcal{N}\big(y_{t-1}; \tilde{\mu}_t(y_t, x_0), \tilde{\beta}_t I\big). \tag{56}$$

Thus, the intermediate posterior of CPDM is a condition-fixed translated version of the standard DDPM posterior.

A.6 Reverse parameterization with η-prediction

In CPDM, the forward stochastic term is not ϵ alone but

$$\eta := \epsilon + r_t(c). \tag{57}$$

The network is trained to predict

$$\hat{\eta}_\theta(x_t, t, c) \approx \eta. \tag{58}$$

Because $r_t(c)$ is deterministic for each fixed condition c, we can recover the standard DDPM noise prediction by

$$\hat{\epsilon}_\theta(x_t, t, c) = \hat{\eta}_\theta(x_t, t, c) - r_t(c). \tag{59}$$

In the translated y-world, the reverse Gaussian can therefore be parameterized in the standard DDPM form:

$$p_\theta^{(y)}(y_{t-1} \mid y_t, c) = \mathcal{N}\left(y_{t-1}; \mu_\theta^{(y)}(y_t, t, c), \sigma_t^2 I\right), \tag{60}$$

with mean

$$\mu_\theta^{(y)}(y_t, t, c) = \frac{1}{\sqrt{\alpha_t}}\left(y_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\epsilon}_\theta(x_t, t, c)\right). \tag{61}$$

Substituting $\hat{\epsilon}_\theta = \hat{\eta}_\theta - r_t(c)$ yields:

$$\mu_\theta^{(y)}(y_t, t, c) = \frac{1}{\sqrt{\alpha_t}}\left(y_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\big(\hat{\eta}_\theta(x_t, t, c) - r_t(c)\big)\right). \tag{62}$$

Returning to the original x-space gives

$$x_{t-1} = y_{t-1} + m_{t-1}(c), \tag{63}$$

and hence

$$p_\theta^{(x)}(x_{t-1} \mid x_t, c) = \mathcal{N}\left(x_{t-1}; \mu_\theta^{(x)}(x_t, t, c), \sigma_t^2 I\right), \tag{64}$$

where

$$\mu_\theta^{(x)}(x_t, t, c) = \mu_\theta^{(y)}(x_t - m_t(c), t, c) + m_{t-1}(c). \tag{65}$$

Therefore, CPDM does not require a new reverse-time Gaussian form. Instead, it reuses the standard DDPM reverse parameterization in the translated y-world and maps the result back to x-space.

A.7 ELBO equivalence

We now show that the CPDM variational objective in x-space is equivalent to the standard DDPM objective in the translated y-world.

The conditional ELBO in x-space can be written as

$$\begin{aligned}\mathcal{L}_x(x_0, c) := \mathbb{E}_{q_x}\Big[-\log p_\theta^{(x)}(x_0 \mid x_1, c)\Big] \\ + \sum_{t=2}^{T} D_{\mathrm{KL}}\Big(q_x(x_{t-1} \mid x_t, x_0, c) \,\|\, p_\theta^{(x)}(x_{t-1} \mid x_t, c)\Big) \\ + D_{\mathrm{KL}}(q_x(x_T \mid x_0, c) \,\|\, p(x_T \mid c))\,.\end{aligned} \tag{66}$$

By Section A.4, the terminal KL term is identical to the standard DDPM terminal KL term in the y-world:

$$D_{\mathrm{KL}}(q_x(x_T \mid x_0, c) \,\|\, p(x_T \mid c)) = D_{\mathrm{KL}}(q_y(y_T \mid x_0) \,\|\, p(y_T))\,. \tag{67}$$

By Section A.5 and the same Jacobian-1 translation argument, each intermediate KL term satisfies

$$\begin{aligned}D_{\mathrm{KL}}\Big(q_x(x_{t-1} \mid x_t, x_0, c) \,\|\, p_\theta^{(x)}(x_{t-1} \mid x_t, c)\Big) \\ = D_{\mathrm{KL}}\Big(q_y(y_{t-1} \mid y_t, x_0) \,\|\, p_\theta^{(y)}(y_{t-1} \mid y_t, c)\Big)\,.\end{aligned} \tag{68}$$

Finally, because $C_0 = 0$ implies $r_0(c) = 0$ and hence $m_0(c) = 0$, we have

$$x_0 = y_0. \tag{69}$$

Therefore the reconstruction term is also unchanged under the translation:

$$\mathbb{E}_{q_x}\Big[-\log p_\theta^{(x)}(x_0 \mid x_1, c)\Big] = \mathbb{E}_{q_y}\Big[-\log p_\theta^{(y)}(x_0 \mid y_1, c)\Big]\,. \tag{70}$$

Combining the three parts, we obtain

$$\mathcal{L}_x(x_0, c) = \mathcal{L}_y(x_0, c), \tag{71}$$

where $\mathcal{L}_y$ is the standard DDPM variational objective in the translated y-world. Hence CPDM preserves the Gaussian tractability and ELBO/KL structure of DDPM under the deterministic condition-fixed translation induced by $m_t(c)$.

A.8 Scope of the proof and relation to the Schrödinger-inspired coordinate system

The proof above relies only on the following facts:

1. for each fixed condition c, the drift $r_t(c)$ is deterministic,
2. the forward perturbation remains Gaussian after translation,
3. the map $x_t \leftrightarrow y_t$ is a bijective translation with unit Jacobian.

Therefore, the KL/ELBO alignment result does not depend on integrating any additional latent-time Schrödinger dynamics. The role of the Schrödinger-inspired spinor-component formalism in this work is instead to provide a bounded and interpretable coordinate construction from which the condition-dependent drift is built.

In this sense, the mathematical contribution of CPDM is not to introduce a new stochastic family beyond DDPM, but to reorganize conditional generation as a deterministic mean-geometry shift in the forward process while preserving the standard DDPM variational machinery in the translated y-world.

B Practical Sampling Algorithm

Algorithm 1 summarizes practical reverse sampling of CPDM in the DDPM-equivalent y-world. At each timestep, the model first predicts η in the original x-space, recovers the corresponding DDPM noise prediction, performs the standard DDPM reverse update in the translated y-world, and maps the result back to the original x-space.

Algorithm 1 Sampling of CPDM in the DDPM-equivalent y-world

Require: Trained network $\hat{\eta}_\theta$, condition c

Sample $y_T \sim \mathcal{N}(0, I)$
Set $x_T = y_T + \sqrt{1 - \bar{\alpha}_T}\, r_T(c)$
for $t = T, T-1, \ldots, 1$ **do**
  Compute $r_t(c)$
  Predict the drift-augmented target:

$$\hat{\eta}_\theta(x_t, t, c)$$

  Recover the DDPM noise prediction:

$$\hat{\epsilon}_\theta = \hat{\eta}_\theta(x_t, t, c) - r_t(c)$$

  Move to the translated world:

$$y_t = x_t - \sqrt{1 - \bar{\alpha}_t}\, r_t(c)$$

  Compute the DDPM mean in the y-world:

$$\mu_\theta(y_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(y_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\epsilon}_\theta \right)$$

  Sample $z \sim \mathcal{N}(0, I)$ if $t > 1$, and set $z = 0$ if $t = 1$
  Update

$$y_{t-1} = \mu_\theta(y_t, t) + \sigma_t z$$

  Return to the original world:

$$x_{t-1} = y_{t-1} + \sqrt{1 - \bar{\alpha}_{t-1}}\, r_{t-1}(c)$$

end for
return x_0

C Training and Evaluation Protocol

C.1 Dataset preprocessing

All images are resized to 128×128 resolution before training and evaluation. For the main Leaf/Flower experiments, we use the Plant Disease Leaf dataset and the Oxford Flowers 102 dataset as the two endpoint domains. This setting is used as the primary benchmark for evaluating whether forward geometry formation improves conditional generation under compact scalar control.

For the additional CelebA-HQ validation experiments reported in the appendix, we use the CelebAHQ-Gender annotation dataset, which provides male/female labels for CelebA-HQ images. The original CelebA-HQ dataset contains 30,000 high-quality face images, and the CelebAHQ-Gender repository provides gender annotations obtained through classifier-based labeling followed by manual inspection. The annotated dataset contains 19,057 female images and 10,943 male images. From this dataset, we sort the image files deterministically and use 10,000 images per gender class to construct a balanced male/female training set.

The same preprocessing pipeline is applied across CPDM and all DDPM-based baselines. This ensures that the comparison focuses on the effect of condition placement and forward-geometry design, rather than differences in data resolution, class balance, or preprocessing.

C.2 Training setup

All models are trained using the Adam optimizer with a learning rate of 1×10^{-4}, global-norm gradient clipping with threshold 3, and a batch size of 32. We use mixed-precision training with a loss-scaled optimizer. Unless otherwise stated, all DDPM-based baselines and CPDM variants share the same denoising backbone. This fixed-backbone setting is used to isolate the effect of the proposed forward drift and conditioning strategy, rather than architectural differences.

For the Leaf/Flower experiments, models are trained up to 50K steps. We evaluate checkpoints from 30K to 50K at 5K-step intervals. For the additional CelebA-HQ validation experiments, models are trained up to 90K steps to allow more detailed intra-domain attribute learning. We evaluate checkpoints from 50K to 90K at 10K-step intervals.

Training was performed on an NVIDIA L4 GPU. The Leaf/Flower experiments required approximately 20 hours per run, while the CelebA-HQ experiments required approximately 35 hours per run. Sampling was performed on an NVIDIA A100 GPU and required approximately 70 minutes per 1,000 generated samples.

C.3 Checkpoint selection

For each model, we first train and evaluate checkpoints using training seed 777. Within the predefined evaluation range, we select the best-performing checkpoint step according to Avg FID over the two endpoint domains. In the Leaf/Flower setting, the checkpoint candidates are selected from 30K, 35K, 40K, 45K, and 50K steps. In the additional CelebA-HQ validation setting, the checkpoint candidates are selected from 50K, 60K, 70K, 80K, and 90K steps.

After the best checkpoint step is identified using seed 777, we train the same model again with an independent training seed, 2026. For this second run, we evaluate the checkpoint at the same selected step rather than reselecting the best checkpoint independently. This protocol tests whether the performance observed at the selected training step is maintained under a different random initialization and training trajectory.

The same checkpoint-selection protocol is applied consistently across CPDM and all baseline models within each dataset setting.

C.4 FID/KID evaluation

We evaluate generation quality using FID and KID. For the main comparison with baseline models, we generate 5,000 samples per domain from each evaluated checkpoint. For ablation studies, we use 1,000 generated samples per domain unless otherwise specified.

To avoid variation caused by repeatedly estimating real-data statistics, we precompute the real-domain feature statistics from the full training set of each domain and store them as NPZ files. During evaluation, generated samples are compared against the corresponding fixed real-domain statistics. This makes the FID computation consistent across different checkpoints, models, and training seeds.

For the main reported results, we compute domain-wise FID/KID for the seed-777 selected checkpoint and for the independently trained seed-2026 model at the same checkpoint step. The final reported scores are averaged across these two training seeds. KID is computed under the same domain-wise evaluation protocol, using the same real/generated domain pairing.

C.5 Seed protocol

We use two independent training seeds, 777 and 2026, to evaluate whether the observed performance is robust to different model initializations and training trajectories. Training seed 777 is used for checkpoint selection: each model is evaluated over the predefined checkpoint range, and the best checkpoint step is selected according to Avg FID over the two endpoint domains.

Training seed 2026 is then used as an independent validation run. The model is trained from a different initialization, and evaluation is performed at the same checkpoint step selected from seed 777. Thus, seed 2026 is not used to select a new best checkpoint, but to test whether the performance at the seed-777 selected step remains stable under an independent training run.

For the Leaf/Flower experiments, this protocol is applied consistently to CPDM and all baseline models in the main comparison. For the additional CelebA-HQ validation experiments, the same protocol is applied to a balanced subset of 10,000 male and 10,000 female images selected by deterministic file sorting. This is especially important for CelebA-HQ, where gender conditioning corresponds to a subtle intra-domain attribute separation rather than a coarse cross-domain separation.

D Time Scheduling and r_t Trajectory

In CPDM, the conditional drift is not applied as a static perturbation independent of time. Instead, its strength follows a time-dependent cumulative trajectory C_t, which determines how strongly the condition-dependent drift is expressed at each diffusion step. For the scalar drift setting, we define

$$r_t(c) = A\, C_t\, s_z(c)\, \hat{u}_z, \tag{72}$$

where A controls the overall drift strength, $s_z(c) \in [-1, 1]$ is the signed condition coordinate, and $\hat{u}_z$ is a fixed image-space drift direction. The schedule $C_t \in [0, 1]$ satisfies $C_0 = 0$ and $C_T = 1$, so that the drift is absent near the clean data endpoint and fully expressed near the terminal noise endpoint.

Equivalently, the drift trajectory can be written in terms of discrete increments

$$\tau_t = C_t - C_{t-1}, \qquad C_t = \sum_{k=0}^{t} \tau_k. \tag{73}$$

Thus, τ_t controls how fast the drift trajectory moves at each timestep, while C_t controls the accumulated drift position. The actual mean displacement in the forward distribution is

$$\sqrt{1 - \bar{\alpha}_t}\, r_t(c) = \sqrt{1 - \bar{\alpha}_t}\, A\, C_t\, s_z(c)\, \hat{u}_z. \tag{74}$$

Therefore, the effective condition separation is determined by both the DDPM noise scale $\sqrt{1 - \bar{\alpha}_t}$ and the drift schedule C_t. For two opposite conditions $s_z = +1$ and $s_z = -1$, the drift-induced mean separation becomes

$$\Delta\mu_t = 2\sqrt{1 - \bar{\alpha}_t}\, A\, C_t\, \hat{u}_z. \tag{75}$$

This shows that the time schedule does not merely change the magnitude of drift, but directly controls the trajectory by which conditional terminal geometry is formed.

D.1 Cosine-τ Schedule

The cosine-τ schedule uses a smooth cumulative trajectory:

$$\tilde{C}_t = \frac{1 - \cos\left(\pi \dfrac{t}{T}\right)}{2}, \tag{76}$$

followed by discrete differencing and normalization:

$$\tau_t = \tilde{C}_t - \tilde{C}_{t-1}, \qquad C_t = \sum_{k=0}^{t} \tau_k. \tag{77}$$

In implementation, a small lower bound τ_0 is applied to nonzero increments before renormalization, preventing extremely small update regions from becoming numerically inactive. This schedule produces a smooth S-shaped cumulative trajectory: the drift changes slowly near the endpoints and more strongly in the middle timesteps.

This design is useful because reverse sampling proceeds from $t = T$ to $t = 0$. Therefore, the early reverse stage begins from a strongly separated terminal geometry, while the late reverse stage gradually reduces the drift effect, allowing the denoiser to focus more on fine-grained image restoration. In this sense, cosine-τ scheduling separates the role of drift into two phases: terminal geometry organization at high-noise steps and detail restoration at low-noise steps.

D.2 Linear Schedule

The linear schedule is defined as

$$C_t = \frac{t}{T}, \qquad \tau_t = C_t - C_{t-1}. \tag{78}$$

Unlike the cosine-τ schedule, the linear schedule changes the accumulated drift at a constant rate. This provides a simple baseline for testing whether the temporal shape of the drift trajectory matters. If linear scheduling performs similarly to cosine-τ, then the main benefit may come from terminal separation itself. If cosine-τ performs better, then the result suggests that the intermediate trajectory, not only the final terminal drift, is important.

D.3 Step-Constant Schedule

The step-constant schedule is defined as

$$C_0 = 0, \qquad C_t = 1 \quad \text{for } t > 0. \tag{79}$$

Equivalently,

$$\tau_1 = 1, \qquad \tau_t = 0 \quad \text{for } t > 1. \tag{80}$$

This schedule immediately applies the full drift after the first timestep and keeps it constant throughout almost the entire trajectory. Therefore, it removes gradual trajectory formation and tests whether simply adding a nonzero terminal drift is sufficient.

Compared with the cosine-τ and linear schedules, the step-constant schedule is a stronger but less flexible intervention. It maximizes condition separation early, but because the full drift remains active even in low-noise regions, it may interfere with fine-detail reconstruction during the late reverse process. Thus, this schedule serves as an ablation for distinguishing terminal separation from temporally controlled trajectory formation.

D.4 Interpretation

The three schedules compare different hypotheses about how conditional drift should be introduced.

Table 2: Summary of drift time schedules and their main roles.

Schedule	C_t **behavior**	**Main role**
Cosine-τ	Smooth nonlinear accumulation	Separates terminal geometry while reducing late-stage interference
Linear	Uniform accumulation	Tests whether a simple gradual trajectory is sufficient
Step-constant	Immediate full drift	Tests whether terminal drift alone is enough

The important distinction is that CPDM does not only introduce a condition-dependent shift $r_T(c)$ at the terminal point. Instead, it defines an entire trajectory $r_t(c)$ over diffusion time. Therefore, different time schedules can produce the same final drift strength $r_T(c)$, while inducing different intermediate paths. This is important because the denoiser is trained and sampled across all timesteps, not only at $t = T$. As a result, the temporal formation of the drift can affect both conditional separation and reconstruction fidelity.

In our experiments, this schedule ablation is used to verify whether the performance gain comes merely from applying a nonzero drift, or from forming a structured forward geometry over time. If the cosine or linear schedule outperforms the step-constant schedule, it suggests that the trajectory by which the conditional geometry is formed matters, rather than only the final magnitude of $r_T(c)$.

D.5 Forward Trajectory under Different Time Schedules

We resize $\hat{u}_z$ to $128 \times 128 \times 3$ and apply a global scaling factor such that its average pixel-wise ℓ_2 norm becomes 0.6. With $A = 2$ and $|s_z| = 1$, we plot the effective mean displacement $\left\|\sqrt{1-\bar{\alpha}_t}\, r_t\right\|_{\text{avg}}$ during reverse sampling. Unlike the raw r_t trajectory, this curve includes the DDPM noise scale $\sqrt{1-\bar{\alpha}_t}$, and therefore directly represents the magnitude of the condition-dependent forward mean shift.

D.6 Performance Comparison by Drift Time Schedule

All settings use the same drift strength $A = 2$ and the same drift direction $\hat{u}_z$, while only the temporal schedule C_t is changed. The reported value is the average FID score computed from the Flower and Leaf domains.

The linear schedule achieves the lowest average FID of **56.72**. The cosine-τ schedule obtains an average FID of **61.80**, while the step-constant schedule shows the worst performance at **72.32**.

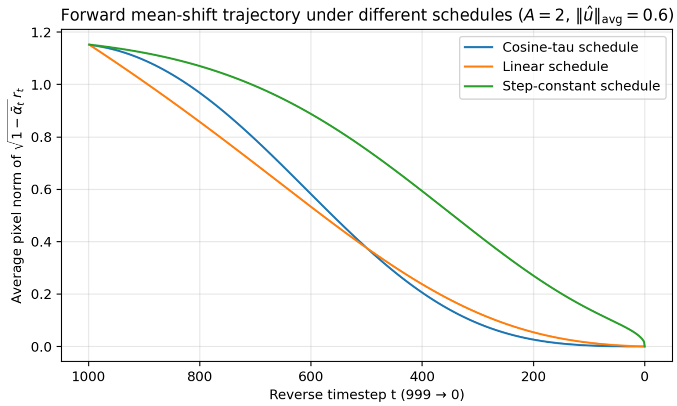


Figure 5: Forward mean-shift trajectory induced by different drift time schedules ($A = 2$, $\|\hat{u}_z\|_{\text{avg}} = 0.6$). The y-axis shows the effective mean displacement $\left\|\sqrt{1-\bar{\alpha}_t}\, r_t\right\|_{\text{avg}}$ during reverse sampling (t=999 $\rightarrow$ 0), which includes the DDPM noise scale and therefore directly represents the magnitude of the condition-dependent forward mean shift.

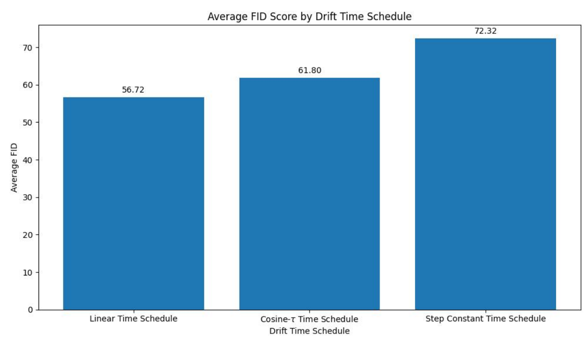


Figure 6: Average FID by drift time schedule (Flower/Leaf). Lower is better. The linear schedule achieves the best average FID (56.72), followed by cosine-τ (61.80). The step-constant schedule performs worst (72.32), suggesting that maintaining the full drift throughout most of the trajectory can degrade reconstruction fidelity.

This result indicates that the final drift magnitude alone is not sufficient to explain the generation quality. Although all schedules reach the same terminal drift strength, they produce different intermediate trajectories during the reverse process. The linear schedule provides the most stable trajectory in this setting, suggesting that a gradual and uniform reduction of the drift effect during denoising better balances conditional separation and image reconstruction.

In contrast, the step-constant schedule applies the full drift for most of the trajectory. This produces strong condition separation, but it can also interfere with fine image restoration in later denoising steps. Therefore, the result supports the view that CPDM benefits not only from terminal geometry separation, but also from the way the drift-induced geometry is formed and relaxed over time.

E Types of $\hat{u}$ basis and their spatial energy characteristics

E.1 Analysis of $\hat{u}_z$ type and FID score

To examine how the choice of drift basis affects forward geometry formation, we compare several types of $\hat{u}$ bases: raw-only, lowpass-only, and their convex combinations. In all cases, the basis is projected to image space at 128×128 resolution and globally rescaled so that the average pixel-wise ℓ_2 norm is fixed at 0.6. This allows the comparison to focus on spatial energy distribution rather than total magnitude.

Table 3: Quantitative comparison of different $\hat{u}$ basis types under a fixed drift strength. All results are computed using 1,000 generated samples per domain, and the best checkpoint is selected by the lowest Avg FID within each dataset setting. Lower FID/KID is better. The comparison evaluates the effect of spatial energy allocation, but does not fully decouple basis design from the optimal choice of $A = 2$.

Dataset	$\hat{u}$ **basis**	**Avg FID** $\downarrow$	**Avg KID** $\downarrow$	**FID gap** $\downarrow$	**Best Step**
Flower/Leaf	Raw only	**51.94**	**.0344**	**11.39**	50K
Flower/Leaf	Lowpass only	67.09	.0465	30.04	40K
Flower/Leaf	$0.7\hat{u}_{raw} + 0.3\hat{u}_{lp}$	56.84	.0381	18.12	50K
Flower/Leaf	$0.3\hat{u}_{raw} + 0.7\hat{u}_{lp}$	69.77	.0484	40.59	40K
CelebA-HQ	Raw only	56.88	.0521	2.75	90K
CelebA-HQ	Lowpass only	55.66	.0530	6.85	70K
CelebA-HQ	$0.7\hat{u}_{raw} + 0.3\hat{u}_{lp}$	**49.07**	**.0437**	**2.43**	70K
CelebA-HQ	$0.3\hat{u}_{raw} + 0.7\hat{u}_{lp}$	54.26	.0500	4.55	70K

Table 3 suggests that mixed bases provide a robust compromise across inter-domain and intra-domain settings. For Flower/Leaf, the raw-only basis performs best, showing that direct prototype differences are effective for coarse domain separation. Nevertheless, the $0.7\hat{u}_{raw} + 0.3\hat{u}_{lp}$ basis is the second-best configuration, indicating that moderate low-pass smoothing preserves most of the useful domain-separating structure. For CelebA-HQ, the same mixed basis achieves the best Avg FID, suggesting that intra-domain facial attribute control benefits from smoothing excessive high-frequency or identity-sensitive drift energy while retaining condition-relevant raw structure. The next subsections analyze this trend through the corresponding heatmap patterns.

E.2 Leaf and Flower $\hat{u_z}$ Heatmap Analysis

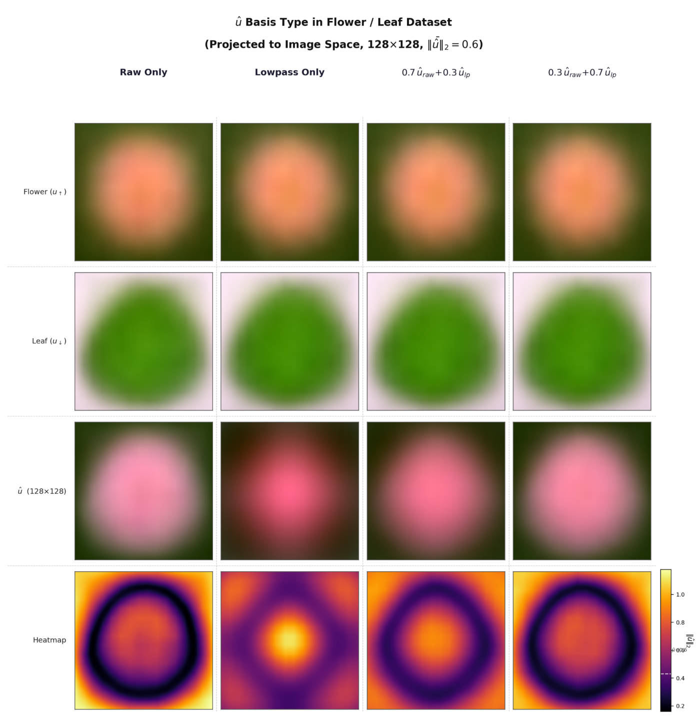


Figure 7: Comparison of $\hat{u}$ basis types in the Flower/Leaf dataset. The raw-only, lowpass-only, and mixed bases are all projected to 128×128 image space and globally normalized to the same average pixel-wise ℓ_2 norm.

The figures show that different basis constructions preserve different geometric properties. The raw-only basis retains more of the original domain difference structure, including sharper spatial contrast and stronger boundary-related variation. The lowpass-only basis suppresses high-frequency fluctuations and instead emphasizes broad, smooth, coarse-scale structure. The mixed bases interpolate between these two extremes, preserving part of the raw structural contrast while maintaining the smoother large-scale organization induced by the lowpass component.

A useful way to interpret these differences is through the heatmaps of $\|\hat{u}_{h,w,:}\|_2$, which reveal where the drift energy is concentrated. In the Flower/Leaf example, the raw-only basis produces a strong ring-like or boundary-oriented energy profile, indicating that the basis emphasizes global object-versus-background contrast and shape boundary information. By contrast, the lowpass-only basis yields a more centrally concentrated energy profile, reflecting smoother coarse object structure. The mixed bases produce intermediate distributions, where coarse object-level geometry remains visible while excessive concentration on a single region is moderated.

E.3 CelebA-HQ Gender $\hat{u_z}$ Heatmap Analysis

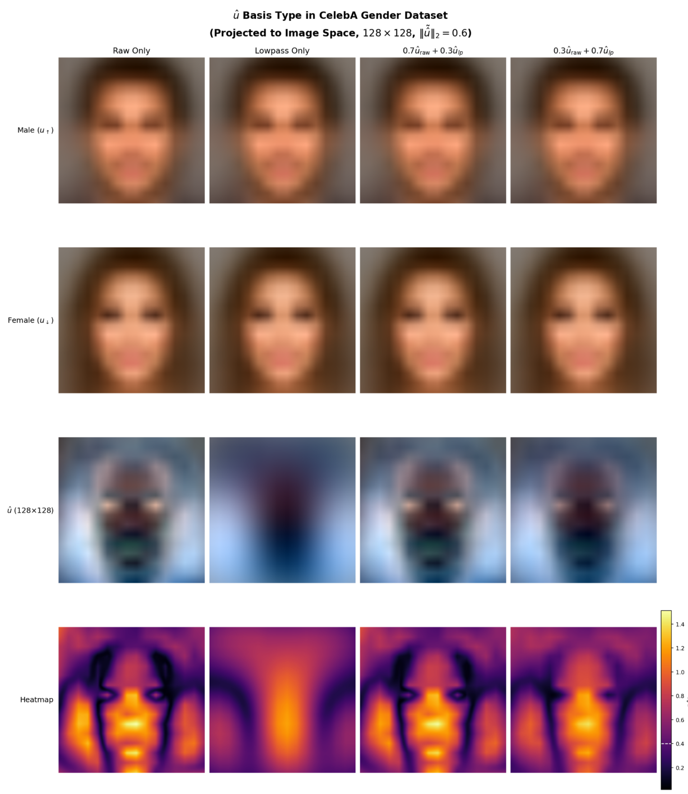


Figure 8: Comparison of $\hat{u}$ basis types in the CelebA-HQ gender dataset. Unlike Flower/Leaf, gender separation is an intra-domain attribute-control problem within the shared face manifold, and the drift energy is distributed over facial contour, hair, and peripheral support regions rather than forming a simple object/background separation.

In the CelebA-HQ gender example, the interpretation is more subtle. Unlike Flower/Leaf, male versus female is not a cross-domain separation between fundamentally different object classes, but rather an attribute-level separation within the same overall face domain. Accordingly, the drift energy tends to be distributed less as a simple object/background contrast and more as an imbalanced spatial pattern over facial contour, hair region, and peripheral facial support regions. In this case, the heatmap often shows relatively stronger energy allocation in outer or side regions, while the central facial region may exhibit comparatively reduced drift energy.

E.4 Comparison with Baselines on CelebA-HQ

Table 4: Quantitative comparison with baselines on CelebA-HQ gender under training seed 777. All results are evaluated using 5,000 generated samples per domain, and the reported checkpoint is selected by the lowest Avg FID over the CelebA-HQ endpoint domains. Lower FID/KID is better.

Model	Avg FID ↓	Avg KID ↓	FID gap ↓	Best Step
Cond. DDPM (one-hot)	46.11	.0474	7.20	90K
Jointly-learned 256D	44.03	.0452	10.21	90K
CPDM ($1.0\hat{u}_{raw}$)	48.34	.0506	0.52	90K
CPDM ($0.7\hat{u}_{raw} + 0.3\hat{u}_{lp}$)	40.83	.0429	3.23	70K
Shift-DDPM++	70.92	.0798	0.55	50K
LLaVA-1.5 + CLIP text emb.	35.50	.0334	8.05	90K
CLIP 512D image emb.	22.14	.0186	4.22	90K

Table 4 reports the CelebA-HQ baseline comparison under the 5K-sample evaluation protocol. The mixed-basis CPDM variant outperforms one-hot conditioning, jointly-learned 256D conditioning, and Shift-DDPM++, showing that forward geometry design can remain competitive in an intra-domain attribute-control setting. However, LLaVA-caption with CLIP text embedding and CLIP image embedding achieve lower FID, indicating that high-capacity semantic conditioning remains stronger for subtle face-attribute control.

This phenomenon is important for interpreting the role of the drift basis. When the task is to separate different semantic domains, a strong central or object-aligned drift may directly support coarse class-level separation. However, when the task is to separate attributes within the same domain, over-concentrating the drift on fine central facial details may be undesirable. Under the $\epsilon + r_t$ target, such a basis may force the forward drift to encode identity-sensitive facial details, which can interfere with reconstruction fidelity or induce overly rigid separation. Instead, a basis whose energy is more concentrated around outer facial or peripheral regions may serve as a coarse geometric bias, while leaving fine central facial details to be resolved by the denoising network through η-target prediction.

In this sense, reduced central energy should not necessarily be interpreted as a weakness of the basis. Rather, it may indicate an appropriate division of labor: the forward drift handles the large-scale conditional geometry, while the reverse denoiser refines the residual fine-grained details. For same-domain attribute separation, such as CelebA-HQ gender, this distinction can be especially relevant because many discriminative cues are subtle and entangled with identity-preserving facial structure. By leaving some of these details as a secondary burden for η-target prediction, the model may achieve a more stable balance between conditional separability and image realism.

Overall, these observations suggest that the design of $\hat{u}$ should be understood not merely as choosing a direction of mean shift, but as selecting a spatial allocation strategy for condition-dependent forward geometry. Raw, lowpass, and mixed bases induce different geometric priors in the forward process, and their suitability may depend on whether the target condition corresponds to coarse cross-domain separation or subtle intra-domain attribute control.

These results also suggest a broader forward-design space beyond the fixed-basis comparison considered here. Because the spatial structure of $\hat{u}$ interacts with the drift strength A and the $\epsilon + r_t$ target, alternative designs such as a smoother common basis with stronger drift strength may form different and potentially more transferable forward geometries. Exploring such basis-strength combinations remains an interesting direction for future work.

F Base CPDM and Continuous CPDM Scalar Traversal

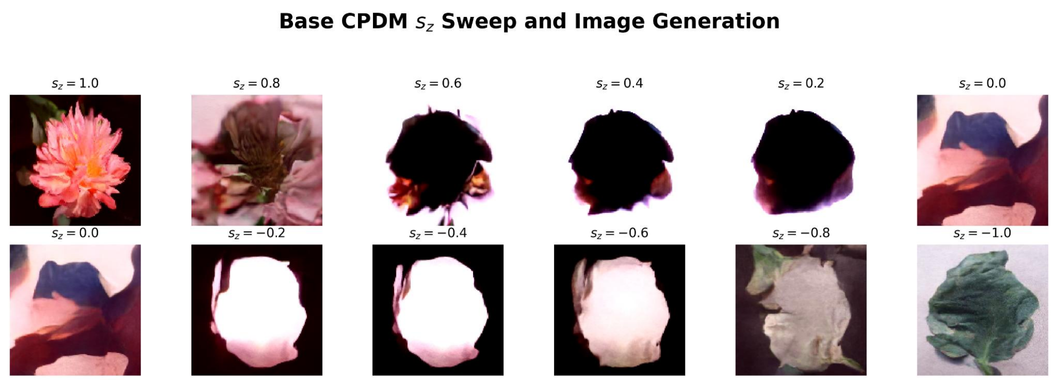
Table 5 reports the endpoint-domain performance of Base CPDM and Continuous CPDM under Training seed 777 at the selected checkpoint (50K steps). This comparison examines how the pairwise consistency objective affects endpoint fidelity and scalar-coordinate traversal behavior.

The endpoint metrics show that the best Continuous CPDM setting introduces only a small endpoint-fidelity trade-off relative to Base CPDM. In particular, $\lambda = 0.1$ slightly increases Avg FID from 43.71

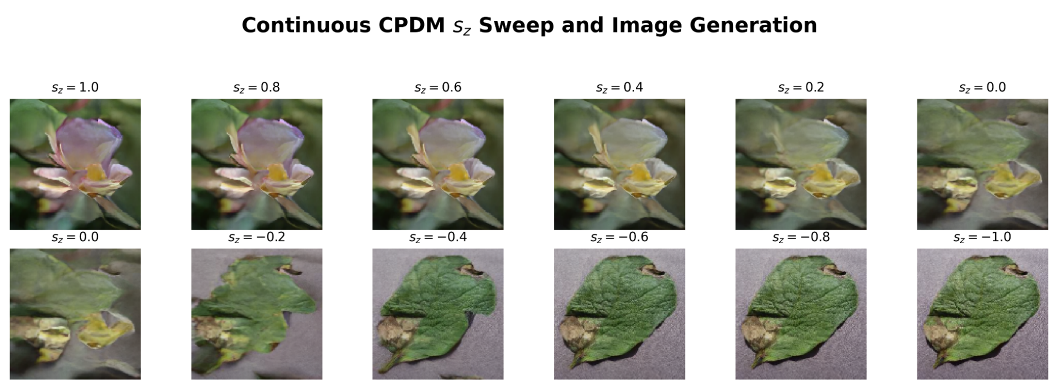
Table 5: Endpoint performance of Base CPDM and Continuous CPDM on the Leaf/Flower setting under Training seed 777. FID/KID are evaluated using 5,000 generated samples per endpoint domain. Lower is better.

Model	Avg FID ↓	Avg KID ↓	FID Gap ↓
Base CPDM	43.71	0.0345	11.60
Continuous CPDM ($\lambda = 0.1$)	45.29	0.0333	17.71
Continuous CPDM ($\lambda = 0.2$)	47.66	0.0367	32.69
Continuous CPDM ($\lambda = 0.3$)	62.08	0.0514	52.47

to 45.29, while maintaining a comparable Avg KID. This suggests that the consistency objective does not primarily improve endpoint generation quality, but instead affects how the model responds to scalar coordinates between the endpoint regions. Figure 9 compares the s_z traversal from $s_z = +1.0$ to $s_z = -1.0$ at intervals of 0.2 under a shared fixed noise seed $y_T \sim \mathcal{N}(0, I)$. Base CPDM remains relatively stable near the endpoint side, including around $|s_z| = 0.8$, but it does not reliably reflect the unseen interior scalar coordinates after inference moves away from the training boundary bands. In contrast, Continuous CPDM produces non-collapsed outputs across the sweep and more consistently reflects changes in the s_z coordinate, indicating that the pairwise consistency objective stabilizes the scalar control axis with only a small endpoint-fidelity trade-off.



(a) Base CPDM



(b) Continuous CPDM

Figure 9: s_z traversal under a fixed noise seed (y_T shared) for Base CPDM (top) and Continuous CPDM (bottom). Panels are ordered from left to right and top to bottom, corresponding to $s_z \in \{+1.0, +0.8, \ldots, -0.8, -1.0\}$. Base CPDM remains plausible near endpoint-side coordinates but becomes less reliable in reflecting unseen interior scalar coordinates, whereas Continuous CPDM maintains non-collapsed outputs that better reflect changes along the s_z axis.

G Training-Step FID Dynamics Across Baselines

This appendix reports checkpoint-wise FID dynamics to support the checkpoint-selection protocol used in the main experiments. For each setting, all models are evaluated over the same predefined checkpoint range, and the best checkpoint reported in the main tables is selected from that common range.

G.1 Flower/Leaf setting: 30K–50K checkpoints

For the Flower/Leaf experiments, we evaluate checkpoints from 30K to 50K training steps at 5K-step intervals. Specifically, the evaluated checkpoints are 30K, 35K, 40K, 45K, and 50K. The same checkpoint range is used for CPDM and all DDPM-based baselines.

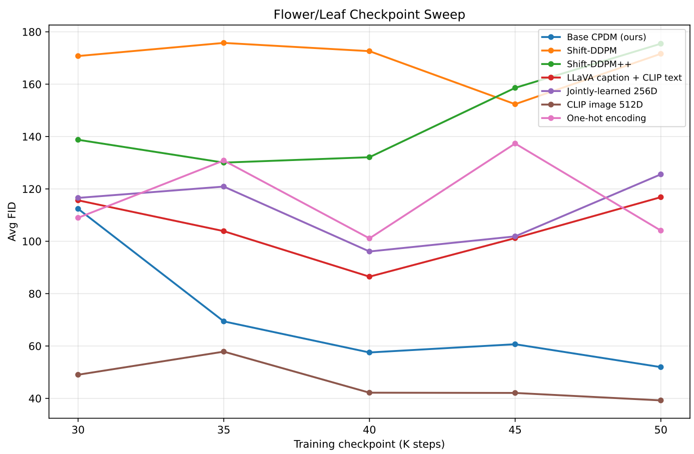


Figure 10: Training-step Avg FID dynamics on the Flower/Leaf setting. All models are evaluated over the same checkpoint range from 30K to 50K steps. Lower FID is better.

The Flower/Leaf dynamics show how each method changes across the common 30K–50K evaluation window. This analysis verifies that the reported best checkpoints are selected under a consistent checkpoint-selection protocol, rather than from model-specific evaluation ranges.

G.2 CelebA-HQ Gender setting: 50K–90K checkpoints

For the CelebA-HQ experiments, we evaluate checkpoints from 50K to 90K training steps at 10K-step intervals. Specifically, the evaluated checkpoints are 50K, 60K, 70K, 80K, and 90K. The same checkpoint range is used for CPDM and all DDPM-based baselines.

These checkpoint-sweep curves are intended to document the common evaluation window and checkpoint-selection protocol, rather than to introduce additional model-specific tuning ranges.

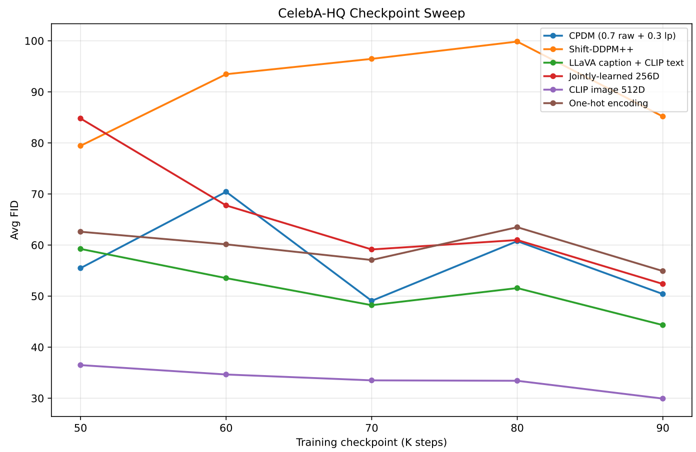


Figure 11: Training-step Avg FID dynamics on the CelebA-HQ Gender setting. All models are evaluated over the same checkpoint range from 50K to 90K steps. Lower FID is better.

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   Answer: [Yes]

   Justification: The paper explicitly separates the general multi-axis formulation from the simplified scalar instantiation actually used in the experiments. The abstract and introduction describe the contribution within this scope, focusing on forward geometry formation, compact scalar conditioning, and continuous generation.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The paper discusses limitations related to the design of the image-space drift direction $\hat{u}$, the difficulty of intra-domain drift-basis selection, and the lack of validation on larger benchmark-scale datasets.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: The appendix provides the assumptions and derivation for the KL/ELBO alignment in the DDPM-equivalent translated y-world. The proof explicitly relies on deterministic condition-fixed drift, Gaussian preservation after translation, and the unit-Jacobian bijective translation.

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   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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   Justification: The paper specifies the datasets, training steps, seed protocol, checkpoint selection procedure, FID/KID evaluation protocol, and fixed evaluation settings for each domain. The reported results use fixed dataset statistics and controlled seed settings to reduce evaluation variability.

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Answer: [No]

Justification: The code and checkpoints are not publicly released at submission time because they are being anonymized and cleaned into a user-friendly reproduction package. We plan to release an anonymized implementation with CPDM and Continuous CPDM checkpoints and reproduction instructions by the end of May, while the paper provides the algorithmic details, datasets, training protocol, and evaluation settings needed to understand the main results.

6. **Experimental setting/details**

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   Justification: The paper reports the dataset construction, image resolution, training steps, learning rate, gradient clipping threshold, sample counts for FID/KID evaluation, checkpoint range, and checkpoint-selection rule. The appendix further provides checkpoint-wise FID dynamics to justify the selected evaluation steps.

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    Answer: [Yes]

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